

HARRISON 20 SSE, NE, USA

WMO: 744380

Lat: 42.4247N

Lon: 103.7363W

Elev: 4406

StdP: 12.50

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 94077

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-11.6	-4.5	-17.3	2.5	-12.1	-10.0	3.8	-4.2	21.0	22.4	18.8	25.1	3.2	N/A	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.7	93.0	62.0	90.0	61.8	87.0	61.7	67.6	82.0	66.3	81.2	64.8	80.1	7.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	101.7	73.2	61.5	96.4	71.6	60.0	91.1	70.5	34.7	81.3	33.5	81.2	32.3	80.3	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.2	14.1	DB	-23.5	98.7	5.4	2.7	-27.4	100.6	-30.5	102.2	-33.5	103.7	-37.4	105.6	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	24.4	25.4	36.7	43.2	52.0	63.5	70.2	67.1	58.2	44.7	33.4	22.8		
	DBStd	18.81	11.44	12.18	10.28	8.83	8.60	6.93	5.49	5.84	7.72	9.38	10.34	12.16		
	HDD50	3791	793	690	420	233	81	2	0	0	17	211	501	843		
	HDD65	7590	1258	1110	876	653	408	107	18	47	228	629	948	1308		
	CDD50	2060	0	0	9	30	143	408	626	531	263	47	3	0		
	CDD65	383	0	0	0	0	5	63	179	113	23	0	0	0		
	CDH74	7238	0	0	8	43	247	1201	2715	2019	893	110	2	0		
	CDH80	2945	0	0	0	5	59	447	1249	852	315	18	0	0		
(14) Wind	WSAvg	6.0	5.7	6.1	6.8	7.6	7.1	6.2	5.3	5.1	5.4	5.8	5.5	5.6		
(15) Precipitation	PrecAvg	14.6	0.2	0.4	0.7	1.7	2.4	2.6	1.8	1.5	1.3	1.2	0.4	0.3		
	PrecMax	19.9	0.6	1.2	2.0	3.3	6.0	5.9	3.0	2.9	3.3	2.7	1.6	0.8		
	PrecMin	8.1	0.0	0.0	0.1	0.2	0.8	0.6	0.6	0.4	0.2	0.1	0.0	0.0		
	PrecStd	2.8	0.2	0.2	0.5	0.9	1.3	1.6	0.7	0.6	0.7	0.7	0.4	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.6	62.5	74.7	78.7	86.0	95.0	98.0	95.9	91.3	82.9	71.5	59.6		
		MCWB	40.6	44.4	49.2	51.4	57.3	61.7	61.9	61.5	61.1	54.8	48.3	42.8		
	2%	DB	51.7	55.6	68.3	73.7	80.8	89.2	93.8	91.6	87.5	77.1	65.4	52.9		
		MCWB	39.2	41.2	46.5	51.3	56.4	61.4	63.4	61.1	59.3	52.7	45.8	39.6		
	5%	DB	46.7	49.9	62.7	68.3	76.0	85.4	90.8	88.7	83.9	71.4	59.4	47.6		
		MCWB	37.0	38.5	44.5	48.8	56.4	61.8	63.2	61.6	57.3	51.1	43.5	36.7		
	10%	DB	42.3	44.0	57.2	62.9	71.0	81.2	87.2	85.2	79.5	65.7	53.3	42.3		
		MCWB	34.8	35.1	42.4	46.5	54.9	61.2	63.5	61.9	56.9	48.6	40.7	34.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	46.1	51.4	54.9	62.4	68.2	69.2	68.6	66.4	57.3	49.1	43.3		
		MCDB	53.9	62.9	72.2	71.7	76.4	82.3	83.9	81.5	84.0	76.0	68.7	58.6		
	2%	WB	39.5	42.1	48.0	52.0	60.0	66.4	68.0	66.8	62.8	54.8	46.2	40.5		
		MCDB	49.5	56.0	67.3	69.0	72.4	79.7	82.7	81.1	78.6	70.8	63.6	53.1		
	5%	WB	37.2	39.7	45.6	49.3	57.9	64.4	66.8	65.1	60.6	52.3	43.8	37.3		
		MCDB	45.9	50.9	62.7	66.0	70.0	78.7	81.9	80.3	76.4	67.0	58.1	47.2		
	10%	WB	34.6	36.1	43.5	47.0	55.7	62.8	65.4	63.7	59.0	50.0	41.1	34.3		
		MCDB	41.9	45.1	57.8	61.2	66.9	77.5	81.2	78.7	74.6	64.1	53.3	42.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	27.4	26.1	28.5	27.3	27.0	29.7	30.7	32.3	34.5	32.1	30.7	28.2		
		MCDBR	32.8	33.1	39.3	40.1	37.2	36.3	37.1	38.7	43.2	44.2	42.0	34.7		
		MCWBR	21.7	18.8	20.4	19.7	15.7	13.8	11.3	13.8	17.8	21.4	23.5	22.5		
	5% WB	MCDBR	31.7	31.3	37.4	36.8	28.2	30.5	29.9	30.5	34.3	36.5	39.6	33.6		
		MCWBR	21.4	18.1	19.6	19.0	14.0	14.0	11.9	12.7	14.7	18.4	22.8	22.0		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.254	0.275	0.304	0.324	0.333	0.361	0.353	0.317	0.279	0.257	0.246		
	taud		2.591	2.559	2.532	2.451	2.464	2.460	2.415	2.419	2.507	2.588	2.566	2.568		
	Ebn at Noon		296	309	311	306	301	297	287	286	292	294	287	285		
	Edh at Noon		20	25	29	34	35	35	36	35	30	24	21	19		
(44) All-Sky Solar Radiation	RadAvg		682	958	1348	1664	1941	2305	2286	1984	1592	1081	749	570		
	RadStd		33	56	102	96	160	140	87	67	94	96	45	35		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page